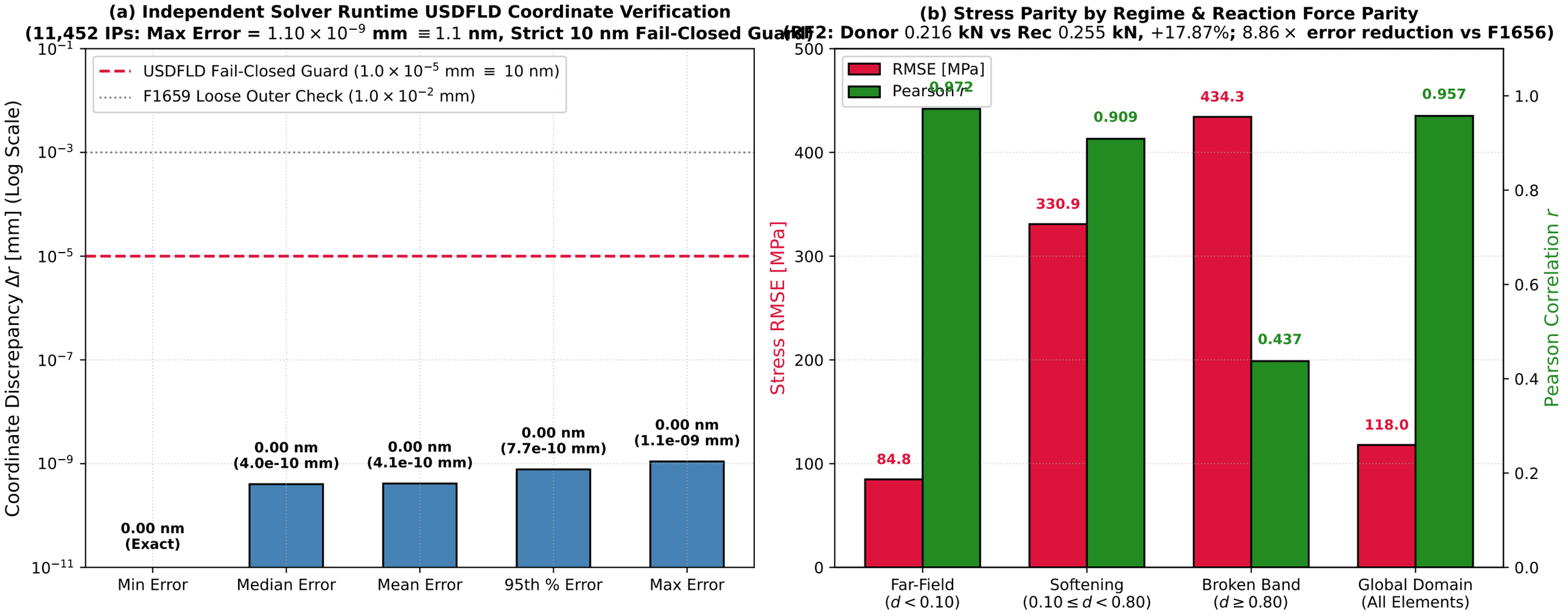
(a) Independent Solver Runtime USDFLD Coordinate Verification
(11,452 IPs: Max Error = 1.10×10^{-9} mm ≡ 1.1 nm, Strict 10 nm Fail-Closed Guard)
USDFLD Fail-Closed Guard (1.0×10^{-5} mm ≡ 10 nm)
F1659 Loose Outer Check (1.0×10^{-2} mm)
Coordinate Discrepancy Δr [mm] (Log Scale)
10^{-1}
10^{-3}
10^{-5}
10^{-7}
10^{-9}
10^{-11}
0.00 nm (Exact)
0.00 nm (4.0e-10 mm)
0.00 nm (4.1e-10 mm)
0.00 nm (7.7e-10 mm)
0.00 nm (1.1e-09 mm)
Min Error
Median Error
Mean Error
95th % Error
Max Error
(b) Stress Parity by Regime & Reaction Force Parity
(RF2: Donor 0.216 kN vs Rec 0.255 kN, +17.87%; 8.86× error reduction vs F1656)
RMSE [MPa]
Pearson r
Stress RMSE [MPa]
Pearson Correlation r
500
400
300
200
100
0
1.0
0.8
0.6
0.4
0.2
0.0
84.8
0.972
330.9
0.909
434.3
0.437
118.0
0.957
Far-Field ($d < 0.10$)
Softening ($0.10 \leq d < 0.80$)
Broken Band ($d \geq 0.80$)
Global Domain (All Elements)